

Week 1: Jumping pegs

Facilitator guide (W1-F-v1)

This guide is the only place with answers, arguments, timings and teaching notes. The student pages contain problems, boards and recording space and nothing else. Please read your own table's section fully before the session; every answer is worked out here so that nothing has to be solved at the table. Section 9 is background for the two mathematicians.

1. The session at a glance

The move. A peg jumps over an orthogonally adjacent peg into the empty square just beyond it, and the jumped peg is removed. That is the only move all hour.

The questions. All three tables meet the same three questions in different sizes: *Can this pattern of pegs be cleared down to one peg? How far past a line can an army of pegs send a single peg? How few pegs does that take?* The headline facts: on a strip, an army of any size sends a peg at most one square past the line, never two; on a grid, armies of 2, 4, 8 and 20 pegs reach the first, second, third and fourth row past the line, and no army, however large, reaches the fifth (Conway's Soldiers). The tool that proves the impossibilities is a value attached to each square so that no jump ever raises the total value of the pegs on the board.

0:00–0:05	Free handling. Pieces and page 1 are on the tables; children do whatever they like with them.
0:05–0:08	Whole-group launch (section 3): the jump, done by everyone on strip A.
0:08–0:50	Table work (sections 5–7). Roughly forty minutes.
0:50–0:58	Share (section 8): each table shows one thing.
0:58–1:00	Pieces back in cups, packets go home.

Who sits where. The K–1 section is written so that the parent can run it: it needs reading aloud, scribing, and the answers below, and nothing has to be worked out. Suggested: parent at K–1, second mathematician at grades 2–3, organizer at grades 4–5, with the organizer stopping at the K–1 table twice (around 0:25 for the river strips and around 0:40 for the boards). Swap the mathematicians if you prefer; both sections are self-contained.

2. Materials and preparation

- Student packets, printed single-sided (children put pieces on the pages and lay two pages side by side): K–1, 11 pages, 4 copies (3 children + 1 spare); grades 2–3, 9 pages, 5 copies; grades 4–5, 9 pages, 4 copies. Three copies of this guide.
- Pieces. Every cell on the K–1 pages is 1 in square, so the K–1 table uses pennies or round counters up to 1 in across. Standard 1.25-inch checkers are too wide: two of them cannot sit in neighbouring squares. Keep the checkers for the launch demonstration. The last K–1 board needs 8 pieces per child, so put 30 pennies or counters in the K–1 cup. (The K–1 pages say “penny”; if you hand out counters, say “counter” when reading aloud.) Grades 2–3 and 4–5 use pennies (cells are 0.8 in; board T at grades 4–5 has 0.75 in cells): 10 per child at grades 2–3, 20 per child at grades 4–5 (the row-four board), plus 30 spare. About 160 pennies in all, sorted into a cup per table.
- One pencil per child. A few blank printed grids (any size) at each table for children who want to invent their own boards.
- For the launch: a whiteboard or large sheet on which a strip of five squares can be drawn, and three magnets or large counters; or simply hold up a K–1 page 1 with pennies on it.
- Preparation, about fifteen minutes: print and sort; each adult plays through the first two problems of their table with real pieces; the K–1 adult also plays boards T, U and W (answers in section 5) so the jumps are familiar.

3. The whole-group launch (two to three minutes)

Children stay at their tables; every child has page 1 and pieces. Organizer:

“Find strip A on your first page and put a piece on each gray square. Here is the only move we use today.” (Show it on the board: three squares, pieces on the first two.) “A piece jumps over the piece right next to it and lands on the empty square just beyond. The piece it jumped over comes off. Do it now on your strip.” Wait until every child has jumped and removed a piece; table adults help.

“Three things are not allowed: jumping over an empty square, landing on a square that already has a piece, and jumping slantwise. Today’s question is this: if a bunch of pieces start on one side of a line, how far past the line can they send one piece? Your table has its own way of finding out.” Start the tables. Do not say anything about answers or about the golden ratio; that story belongs to the share.

4. Running a table

Every sequence on the pages is *act, record, compare, explain*. Let the children do and record before you ask why. When a child says “it is impossible”, the question to ask, once the recording space is filled, is “How do you know for sure?” Accept explanations a child can produce: pointing at a filled table, listing every case, describing a strategy, showing one counterexample, or saying why the same reason works again on a bigger strip. Do not ask for a rule with a letter in it.

Two habits to enforce gently all hour: the jumped piece really comes off, and pieces are reset to the gray squares before a new try. When a problem asks for “the farthest square” or “the highest row”, the record is the best result over all tries, not the last try; on the river strips, “0” means that no peg crossed at all, and that is a real result to record.

5. Table K–1 (two kindergartners and a first grader; adult reads aloud and scribes)

Order and timing. Problems 1–2 (0:08–0:13), 3 (0:13–0:21), 4–5 (0:21–0:30), 6–8 (0:30–0:38), 9–11 (0:38–0:50). Read each problem aloud in pieces: one sentence, then the action. Write numbers in boxes for children who ask. Every problem has a real answer and the children should feel that they found it, so hold hints until a child has tried twice. The pages use letters for strips and boards (A to W); this guide adds column letters and row numbers to the boards for its own descriptions, and the K–1 pages have none.

Problem 1. Strip A (pieces on squares 2, 3): the piece on 2 jumps over 3 and lands on 4, or the piece on 3 jumps over 2 and lands on 1; either is a correct jump, and the jumped piece comes off. Strip B (pieces on 3, 4): the leftward jump is 4 over 3, landing on 2.

Watch for. Sliding instead of jumping; jumping over an empty square; forgetting to remove the jumped piece. Ask: “Which piece jumped? Which piece came off? Where did it land?”

Problem 2. Strip C (pieces on 2 and 4): NO, there is no piece right next to another one. Strip D (all five full): NO, there is no empty square to land on. Strip E (pieces on 2, 3, 4): YES, but only the middle piece can jump (over 3 to 5, or over 2 to 1); each end piece would have to land on an occupied square. Ask which piece can jump and why the end pieces cannot.

Problem 3 (pages 2 and 3). Strips are six squares long, numbered from the left. Each strip has a box for the first try and a box for the fewest pieces left after more tries.

- Strip F (2, 3): always 1 left. First try 1, fewest 1.

- Strip G (2, 3, 4): always 2 left. Only the middle piece can jump; after it does, the two remaining pieces are far apart and stuck.
- Strip H (2, 3, 5): 1 or 2, and the order matters. Jump 2 over 3 to 4, then 4 over 5 to 6 (or 5 over 4 to 3): one left. But if the first jump is 3 over 2 to 1, the pieces on 1 and 5 are stuck: two left. Fewest 1.
- Strip I (1, 2, 4, 5): always 2 left, whatever the order.
- Strip J (1, 3, 5): no jump is possible at all; 3 left on every try.
- Strip K (1, 2, 4, 6): 1 or 2. The only first jump is 1 over 2 to 3. Then 3 over 4 to 5 followed by 6 over 5 to 4 leaves one piece; but 4 over 3 to 2 instead leaves pieces on 2 and 6, stuck. Fewest 1.

The two boxes side by side are the point: on F, G, I and J the two boxes agree whatever the child does, on H and K the first box depends on the order and the second box is 1. Say this aloud with the pages in front of the children: “On these strips the order never mattered. On these two it did.”

Watch for. Children who stop while a jump is still possible. Ask “Is any piece next to another piece with an empty square beyond?”

Hints, in order. For a second try on H or K: “Try starting with the other piece.” “Which jump left two pieces far apart? Can you avoid it?” For strip K after the forced first jump: “Now you have pieces on 3, 4, 6. Two jumps are possible. Try both.”

Problem 4. Strips have four home squares (1–4), the river, then square 1 and square 2 across. Strip L (pieces on 3, 4): 3 jumps over 4 and lands on square 1 across; X there. Strip M (2, 3, 4): YES; only the middle piece can move; 3 over 4 lands on square 1 across, or 3 over 2 lands on home square 1; then everything is stuck. Farthest: square 1. Strip N (1, 2, 3): NO. The only jump is 2 over 3 to home square 4; then the pieces on 1 and 4 are stuck and nothing has crossed. Strip O (1, 2, 3, 4): YES, and the play is forced. The only possible first jump is 3 over 4 onto square 1 across; the only possible second jump is 1 over 2 to 3; then the pieces on home 3 and on square 1 across are two apart and stuck. Farthest: square 1.

Why strip N fails, in words the children can check with the pieces: a piece crosses the river by jumping over a piece on home square 4. On strip N that square is empty, and the only way to fill it (2 over 3 to 4) empties the two squares behind it, so no piece is left to jump over it. Compare with strip M, where square 4 is full from the start.

Watch for. A child who wants to jump from square 1 across back over home square 4 (it is empty after the crossing, so that is not a jump).

Problem 5. Strips P, Q, R: whatever squares the child chooses, no piece ever reaches the star. Circle 1. Two things to notice with the children, strips side by side:

- Some armies never cross at all. An army crosses only if home square 4 is full from the start (as in strip N, a piece that jumps into square 4 empties the two squares behind it, and no piece is left to jump over it). Of the fifteen ways to choose up to four home squares, exactly five cross (3 4; 2 3 4; 1 3 4; 1 2 4; 1 2 3 4) and ten never cross: every single piece, every pair except 3 4, and 1 2 3. So a NO on a free strip is normal; ask the child to look at square 4.
- No army reaches the star. The moment a piece lands on square 1 across, home squares 3 and 4 are empty, and to land on the star a piece would need home square 4 full *and* square 1 across full at the same moment. Refilling home square 4 is itself a jump, which uses up two more pieces behind it and leaves two more empty squares. With four pieces in a block the play was forced (strip O), so that list of cases is complete; for a placement with gaps, let the child exhaust it. (Every choice of home squares was checked by computer; the proof for strips of any length is in section 6, Problem 8, and section 9.)

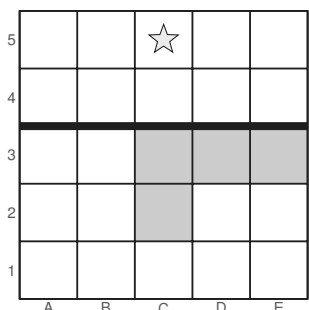
Hints, in order. “Is the square right next to the river full?” “What do the two squares just before the river look like right after a crossing?” “What would a piece need in order to land on the star?”

Problem 6. Board S: the piece on the lower gray square jumps over the upper one and lands in row 1 across, directly above. X there.

Problem 7. Board T (the two-by-two block): the highest square reached is in row 1 across, never the star. In the

natural line each column sends a piece up once, the two crossed pieces then sit side by side in row 1, and they can only jump sideways along that row. Other lines start with a sideways jump inside the block; in every line the highest square reached is in row 1 across. Let the children exhaust it.

Problem 8. Board U (the L: two pieces stacked under the star's column, two more along the river to the right). Three jumps reach the star: the lower stacked piece jumps up over the upper one into row 1; the far right piece jumps left over its neighbor into the square just vacated below the river; that piece jumps up over the row-1 piece onto the star.



Jumps on board U: C2 over C3 to C4; E3 over D3 to C3; C3 over C4 to C5 (the star). (Letters and numbers are for this guide; the K-1 page has none.)

Hints, in order. “Which piece will be the last to move, and what must be right under the star just before?” “Get one piece across first. Now the square under it is empty; can something jump into it from the side?”

Problem 9. Board V, three pieces: they never reach the star; the best is row 1 across (two stacked pieces cross, the third cannot help). Complete reason, which a first grader can follow with the pieces: to land on the star you need, at one moment, a piece in row 1 under the star and a piece on the home square directly below it. Each crossing uses up two pieces, so with three pieces there is only one crossing, and the piece in row 1 got there by jumping straight up, which emptied the home square under it; the third piece is not on that square and cannot jump into it alone, because a jump needs a second piece to hop over. (A piece can also enter row 1 sideways, but that needs two pieces already across, which is four pieces in all.) Section 6, Problem 14 gives the values version: three home squares are worth at most 18 and the star is worth 21.

Hints, in order. “Get one piece across first. Which square must be full under it for the last jump, and can the third piece get there?”

Problem 10. Among all ways of putting four pieces on the fifteen home squares of board V, exactly two reach the star: the L of board U and its mirror image (two stacked under the star, two more along the river to the *left*). This was checked by computer for all 1365 placements. Four other shapes reach row 2 but not the star: the same L with its stack under a different column, for example pieces on B3, C3, D3 and D2 in the guide's letters (two along the river to the left of the stack), which send a piece to D5, the row-2 square just right of the star; likewise A3 B3 C3 with A2 reaches A5, B3 C3 D3 with B2 reaches B5, and C3 D3 E3 with E2 reaches E5. If a child finds one of these, they have found the shape: ask them to slide it along so that the stack is under the star. Every other four-piece shape stops in row 1 or lower: the block never crosses twice in one column; a row of four along the river cannot cross at all (crossing needs two pieces stacked); a T shape crosses once and is stuck.

Hints, in order. “Where were the two stacked pieces in board U? Try putting the other two on the other side.” “Your piece reached row 2 next to the star. Move the whole shape one square over.”

Problem 11. Board W, eight pieces in a T: five along the river (call them left end, left, middle, right, right end) and three under the middle one (upper, middle, lower). Seven jumps reach the big star in row 3:

1. the upper piece of the column jumps up over the middle river piece into row 1;
2. the lower piece of the column jumps up over the column's middle piece;
3. the left end piece jumps over its neighbor into the middle river square;
4. that middle river piece jumps up over the row-1 piece into row 2;
5. the right end piece jumps over its neighbor into the middle river square;
6. the piece just under it jumps up over it into row 1;
7. the row-1 piece jumps over the row-2 piece onto the big star.

In the letters of the guide (columns A–E, rows 1–7, star at C7): C3 over C4 to C5; C1 over C2 to C3; A4 over B4 to C4; C4 over C5 to C6; E4 over D4 to C4; C3 over C4 to C5; C5 over C6 to C7. Do it once yourself beforehand. If the children reach only row 2, that is a real result; show the last three jumps and let them redo the whole thing.

If the table stalls. Reading is the bottleneck, not the mathematics: shorten by reading only the action sentences. If Problem 3 drags, skip page 3 (strips I, J, K) and go to the river (Problem 4), which is the heart of the hour. If Problems 4–5 drag, use only strip P of the three free strips. Board W (Problem 11) is the goal of the hour; keep ten minutes for it. If a child is done listening, hand them a blank grid and ask for “a strip pattern of four pieces that clears to one piece” (for example pieces on squares 1, 2, 4, 6; test any answer by playing it).

If the table finishes early. (a) Comb strips: draw a long strip and put pieces on squares 1, 2, 4, 6, 8, 10, . . . ; it can always be cleared to one piece: jump 1 over 2 to 3, so that 3 and 4 form a pair; jump 3 over 4 to 5, so that 5 and 6 form a pair; and so on to the last tooth, where the final pair collapses to one piece. Ask “add one more tooth, does it still work?” (b) A tall column two squares wide: no matter how deep, no piece ever gets past row 1. On a strip of grid exactly two squares wide (cut one from a blank grid) the children can see why: no sideways jump is possible, so each column is a strip on its own. On a wide grid the pieces can also spread sideways, and the answer is still row 1 (checked by search for depths up to 7, and proved for every depth in section 9); let them try a 2-by-4 column anywhere on a blank grid. (c) A three-by-three block of nine pieces reaches row 2, but only above a side column of the block, never above its middle column; a three-wide block four deep (twelve pieces) reaches row 3 above a side column. Both facts were checked by computer; let them search.

6. Table grades 2–3 (four third graders)

Order and timing. Problems 1–2 (0:08–0:12), 3–4 (0:12–0:19), 5–6 (0:19–0:25), 7–9 (0:25–0:31), 10–12 (0:31–0:38), 13–14 (0:38–0:44), 15 (0:44–0:50). Problem 16 is for a group that finishes early. Board W (Problem 15) is the most satisfying thing in the packet; keep six minutes for it even if that means cutting Problem 14 to its first two blanks.

Problem 1. Strip A (pegs on 3, 4): 3 over 4 lands on 5, or 4 over 3 lands on 2.

Problem 2. Strip B (3 and 5): not possible, no two pegs are adjacent. Strip C (all eight): not possible, no empty landing square. Strip D (3, 4, 5): possible; only the middle peg can jump, landing on 2 or on 6, so X on squares 2 and 6.

Problem 3 (pages 1 and 2). Eight-square strips, numbered from the left.

- Strip E (2, 3, 5): 1 is possible (2 over 3 to 4, then 4 over 5 to 6 or 5 over 4 to 3); starting with 3 over 2 to 1 strands pegs on 1 and 5, leaving 2. Fewest: 1.
- Strip F (2, 3, 5, 6): always 2, in every order (there are four first jumps; try them all with the children if they doubt it).
- Strip G (1, 2, 4, 6): 1 is possible: 1 over 2 to 3, 3 over 4 to 5, then 6 over 5 to 4 (or 5 over 6 to 7). Choosing 4 over 3 to 2 as the second jump leaves 2 and 6 stuck. Fewest: 1.
- Strip H (2–6, five in a row): always 3; the best possible is 3.

A certificate the mathematician can use for strip F, after the children have exhausted it: color the squares 1, 2, 3, 1, 2, 3, . . . from the left. Every jump changes the number of pegs on *each* of the three colors by one (two colors lose a peg, the third gains one), so the three parities all flip together. A single peg has one odd color and two even ones. Strip F starts with no peg of color 1, two of color 2 and two of color 3 (all even); flipping all three parities never produces one odd and two even, so one peg is unreachable. Strip H has parities one odd, two even and still gets stuck at 3: the certificate can prove impossibility but never possibility.

Problem 4. Five-peg patterns that clear to one peg (all of them, on eight squares, with positions from the left): 1-2-3-4-6, 1-2-4-6-8, 1-2-4-7-8, 1-2-5-7-8, 1-3-4-5-6, 1-3-5-7-8, 2-3-4-5-7, 2-4-5-6-7, 3-4-5-6-8, 3-5-6-7-8. Patterns that always end with three or more: five in a row anywhere (3 left), 1-2-3-6-8 (4 left), 1-3-6-7-8 (4 left), and others

(22 of the 56 patterns); any child's pattern is checked by playing it out. Strips I and J are for the clearing pattern (first and second attempt), K and L for the stuck pattern.

Hints, in order. “Start from strip G (1, 2, 4, 6), which clears to one, and add one peg; test where it can go” (square 8 works, so does square 3). For the stuck pattern: “Five in a row got stuck on strip H; does it matter where the row is?”

Problem 5. Six home squares, the river, then squares 1 and 2 across. Strip M (5, 6): 5 over 6 lands on square 1 across. Strip N (4, 5, 6): only 5 can jump, onto square 1 across (or back to 3); farthest 1. Strip O (3–6): four lines of play; three cross to square 1 (5 over 6 first, then 3 over 4 to 5 or 4 over 3 to 2; or 4 over 3 to 2 first, then 5 over 6), one never crosses (4 over 3 to 2, then 6 over 5 to 4); the farthest square is 1. Strip P (1, 2, 3, 4): the play is forced and never crosses: 3 over 4 to 5, then 1 over 2 to 3, and the pegs on 3 and 5 are stuck; farthest 0. Strip Q (1–6): the play is forced: 5 over 6 onto square 1 across, then 3 over 4 to 5, then 1 over 2 to 3; pegs on 3, 5 and square 1 across, all two apart, stuck. Farthest 1.

The rule behind strip P, for the children in their own words: an army crosses only if square 6, the square touching the river, is full from the start. A peg that jumps into square 6 empties squares 4 and 5 behind it, and to cross, square 5 must be full again while 6 is still full; refilling 5 needs a jump from 3 over 4, which needs 4 full again first, which needs a jump from 2 over 3, and so on: each refill empties two more squares behind, and the army runs out. So 1 2 3 4, and even 1 2 3 4 5, never cross, while 3 4 6 does. (After Problem 7 the values say it in one line: with square 6 empty the army is worth at most $1 + 1 + 2 + 3 + 5 = 12$, and square 1 across is worth 13.) A good follow-up for a fast table: “3 4 6 crosses, 2 3 6 does not; why?” (3 over 4 refills square 5 while 6 is still full; from 2 3 6 the only jumps are 2 over 3 to 4 or 3 over 2 to 1, and then nothing is next to 6.)

Problem 6. Strips R1–R3: farthest square 1 if the army crosses at all, never square 2. Of the 63 ways to choose up to six home squares, 21 cross (exactly those with square 6 full and some way to bring a peg next to it) and 42 never cross, among them 2 3 4 5, 1 2 3 4 and 1 2 3 4 5. A child whose three strips all say 0 should be asked to put a peg on square 6 and one on square 5. (Computer check over every subset of the six home squares; section 9 has the proof for strips of any length.)

Problem 7. Values on strip S: 1, 1, 2, 3, 5, 8 on the home squares, 13 on square 1, 21 on square 2. The strip has no gray squares; each table row says where the pennies go.

pegs on	before	after	
all six home squares; the only jump (5 over 8 onto 13)	20	20	same
2 and 3; left (3 over 2 onto 1)	5	1	down
2 and 3; right (2 over 3 onto 5)	5	5	same
5 and 8; right, across (5 over 8 onto 13)	13	13	same
8 and 13; left (13 over 8 onto 5)	21	5	down

Ask, after the table is filled: “When did the total stay the same? When did it drop? Did it ever go up?” The pattern: a jump to the right (toward the star) removes two neighboring values and lands on the next value, which is exactly their sum, so the total is unchanged; a jump to the left lands on a value no bigger than the smaller of the two removed values, so the total drops.

Problem 8. Total of the home squares 20; square 2 is worth 21. The answer is no, and the argument, in the children's terms: the total starts at 20; no jump ever makes the total bigger (rightward jumps keep it, leftward jumps lower it); if a peg ever stood on square 2 the total would be at least 21; so it never does. Check that every child can say why “at least 21” (the star alone is 21 and all other values are positive). This is a complete proof for strip S, whatever the starting squares (fewer pegs give a smaller total).

Problem 9. Strip T: 1, 1, 2, 3, 5, 8, 13 on the seven home squares, 21 on square 1, 34 on square 2. Home total 33, square 2 is 34. No, for the same reason. The children may notice that the home total is always exactly one less than the star (12 and 13 on a strip with five home squares; 20 and 21; 33 and 34); this is a property of these numbers (section 9) and worth saying aloud as “the army is always one short”.

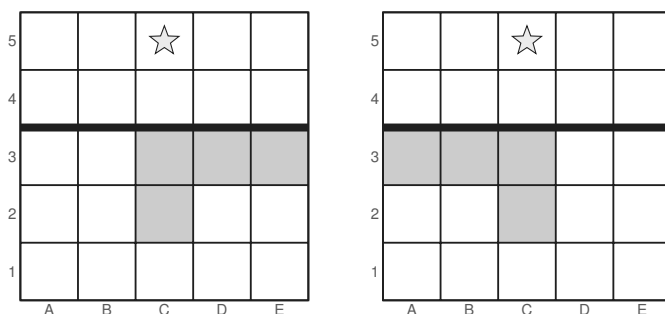
Hints, in order. “The first two values are 1 and 1; what comes next?” “Add the last two values you wrote.”

Problem 10. Board U: C2 over C3 lands on C4.

Problem 11. Three pegs never reach C5; the highest row any peg reaches is 4 (for instance from C2, C3 and anything). The reasoning is the same as at K–1, Problem 9 (one crossing uses two pegs and empties the square under the crossed peg; the third peg cannot refill it alone), and the values reasoning comes in Problem 14. Board U is on page 5, the page before the recording boards.

Hints, in order. “Get one peg across first. Which square must be full under it for the last jump, and can the third peg get there?”

Problem 12. Only shape 4 works: C2 over C3 to C4; E3 over D3 to C3; C3 over C4 to C5. Shape 1 (block) reaches row 4 only; shape 2 (row of four along the river) cannot cross at all; shape 3 (the T) crosses once to row 4 and is stuck. Among all 1365 ways to place four pegs on the fifteen home squares, only shape 4 and its mirror image (C2, C3 with A3, B3) reach the star. Shape 4 or its mirror image shifted one column, for example B3 C3 D3 D2, reaches D5: row 5, but not the star; a child who tries such a shape on the empty squares has found the idea and only needs to slide it under the star.



Hints, in order. “Which square must hold a peg right before the last jump, and which one below it?” “After a peg crosses, refill the square below it from the side.”

Problem 13. Board V values (rows 5 down to 1): 8 13 21 13 8 / 5 8 13 8 5 / 3 5 8 5 3 / 2 3 5 3 2 / 1 2 3 2 1. (a) C2 over C3 to C4: 5+8 before, 13 after, same. (b) E3 over D3 to C3: 3+5 before, 8 after, same. (c) B3 over C3 to D3: 5+8 before, 5 after, down. (d) C4 over C3 to C2: 13+8 before, 5 after, down. Rule on the board: the value of a square is decided by its distance from the star counted in steps along rows and columns; a jump that ends closer to the star keeps the total, any other jump lowers it.

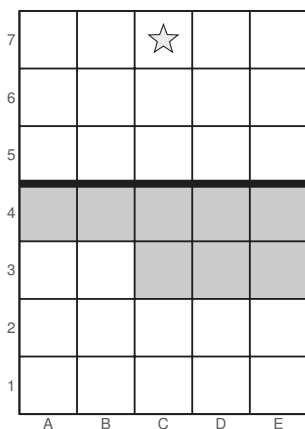
Problem 14. Three most valuable home squares: $8+5+5 = 18$. Star: 21. Shape 1 (C3, D3, C2, D2): $8+5+5+3 = 21$. Shape 4 (C3, C2, D3, E3): $8+5+5+3 = 21$. So three pegs, worth at most 18, can never reach a square worth 21 (the total never rises): this settles Problem 11 for every placement at once. Shape 1 has enough value and still fails: enough value is necessary, not sufficient; the values can only prove that something is impossible, never that it is possible. Shape 4 spends its 21 exactly: each of its three jumps keeps the total, and the lone peg on the star is worth 21.

Problem 15. Board W (T of eight): C3 over C4 to C5; C1 over C2 to C3; A4 over B4 to C4; C4 over C5 to C6; E4 over D4 to C4; C3 over C4 to C5; C5 over C6 to C7. Values (section 7, Problem 12): the eight squares are worth exactly 55, the star’s value, so every jump in a solution must keep the total and the solution ends with a single peg.

Hints, in order. “What must be on C6 and C5 just before the end?” “Get a peg to C5 first, then refill C4 from the side, then jump C4 over C5.” “Now C5 is empty again and C6 is full: you need another peg on C5 with a peg on C4 under it.”

Problem 16. The only other eight-peg sets on board X that reach C7 are the two “flag” shapes: A4 B4 C4 D4 E4 with C3 D3 E3, and its mirror image A4 B4 C4 D4 E4 with A3 B3 C3 (checked by computer over all 125,970 eight-square sets: any set worth at least 55 must contain C4, B4, D4, C3, and then either four of the 5-valued squares, none of which works, or three of them and one of the five 3-valued home squares of the board, giving 50

sets of which exactly the T and these two work). The four small boards are for the sets tried on the way; a failed set with “highest row 6” is worth looking at together.



Right flag: E3 over E4 to E5; D3 over D4 to D5; E5 over D5 to C5; C4 over C5 to C6; A4 over B4 to C4; C3 over C4 to C5; C5 over C6 to C7.

Left flag (mirror image): C3 over C4 to C5; E4 over D4 to C4; C4 over C5 to C6; B3 over B4 to B5; A3 over A4 to A5; A5 over B5 to C5; C5 over C6 to C7.

Hints, in order. “Keep the five pegs along the river. Move the three pegs below the middle to a row along row 3, on one side (C3 D3 E3 or A3 B3 C3).” “Two pegs from that row cross side by side and then one jumps over the other into C5.”

If the table stalls. Problem 3 can absorb a whole hour; after seven minutes move on even if strip H is unfinished. If the values table (Problem 7) meets resistance, do the rows together with pennies on the strip, saying each sum aloud (“5 and 8 go away, 13 arrives”). If Problem 15 is too long, give jumps 1–3 and let them find 4–7.

If the table finishes early. (a) Ask for the values on a strip with nine home squares and whether the army is again one short (1, 1, 2, 3, 5, 8, 13, 21, 34 sum to 88; square 2 is worth 89). (b) Give board T from the grades 4–5 packet (twenty pegs, star in row 4; solution in section 7, Problem 13). (c) Blocks: a two-wide column of any depth never passes row 1 (on a strip of grid exactly two squares wide no sideways jump exists and each column is a strip; on a wide grid the pegs can spread sideways and the answer is still row 1, by search for depths up to 7 and by the proof in section 9); a three-by-three block reaches row 2 above a side column; five pegs along the river with five more under them (five wide, two deep) reach row 3; a five-by-four block of twenty pegs reaches only row 3. All checked by computer.

7. Table grades 4–5 (two fourth graders and a fifth grader)

Order and timing. Problems 1–3 (0:08–0:14), 4–5 (0:14–0:21), 6–8 (0:21–0:29), 9–10 (0:29–0:33), 11–12 (0:33–0:40), 13 (0:40–0:50). Board T (Problem 13, row four, nineteen jumps) is the closing problem and should get its full ten minutes; Problem 14 (a second eight-peg army) is for a group that finishes, and it is a needle-in-a-haystack search without the hint below.

Problem 1. Strip A (pegs on 5, 6 of the six home squares): 5 over 6 lands on square 1 across.

Problem 2. Strip B (4, 5, 6): only 5 can jump; farthest square 1. Strip C (3–6): four lines of play; three cross to square 1, one never crosses (4 over 3 to 2, then 6 over 5 to 4); farthest 1. Strip D (1, 2, 3, 4): forced play that never crosses: 3 over 4 to 5, 1 over 2 to 3, stuck with pegs on 3 and 5; farthest 0. Strip E (1–6): forced play, 5 over 6 across, 3 over 4 to 5, 1 over 2 to 3, stuck; farthest 1.

The rule behind strip D: an army crosses only if square 6, the square touching the river, is full from the start. A peg that jumps into square 6 empties squares 4 and 5 behind it; to cross, square 5 must be refilled while 6 is still full, which needs 4 refilled first, which needs a jump from 2 over 3, and so on, each refill emptying two more squares behind it, until the army runs out. So 1 2 3 4 and even 1 2 3 4 5 never cross, while 3 4 6 does. After Problem 4 the same fact is one line: with square 6 empty the army is worth at most $1 + 1 + 2 + 3 + 5 = 12$ and square 1 across is

worth 13. A follow-up for a fast group: “3 4 6 crosses, 2 3 6 does not; why?” (3 over 4 refills square 5 while 6 is still full.)

Problem 3. Strips F1–F3: farthest square 1 whenever the army crosses at all, never square 2; 21 of the 63 ways to choose up to six home squares cross and 42 do not (every subset was checked by computer; the proof for any strip is Problems 4–5 and section 9). If all three strips read 0, square 6 was empty each time or nothing could reach square 5 while it was full; suggest pegs on 5 and 6 plus any others.

Problem 4. Values 1, 1, 2, 3, 5, 8 on the home squares, 13 on square 1, 21 on square 2. The strip has no gray squares; each row says where the pennies go.

pegs on	before	after	
all six home squares; the only jump (5 over 8 onto 13)	20	20	same
2 and 3; left (3 over 2 onto 1)	5	1	down
2 and 3; right (2 over 3 onto 5)	5	5	same
5 and 8; right, across (5 over 8 onto 13)	13	13	same
the second 1 and 2; left (2 over 1 onto the first 1)	3	1	down
8 and 13; left (13 over 8 onto 5)	21	5	down

After the table: a jump toward the star removes two neighboring values and lands on the next one, which is their sum (the values are built that way), so the total is unchanged; a jump away from the star lands on a value no bigger than the smaller removed value (in the fifth row it lands on 1, equal to the removed 1), so the total drops. The total never goes up.

Problem 5. Home total 20, square 2 worth 21. Proof for strip G: the total starts at 20 (or less, with fewer pegs); no jump raises it; a peg on square 2 would make it at least 21. Strip H: 1, 1, 2, 3, 5, 8, 13 / 21 / 34; home total 33, square 2 worth 34; the same argument word for word. Ask “Why is the army always exactly one short?” and let them test $1+1+2+3+5 = 12$ against 13 and $1+1+2+3+5+8+13+21 = 54$ against 55: adding each Fibonacci number to the sum of all earlier ones gives the next one minus 1 (section 9). Ask also why the same reason works for a strip of any length: the values can be continued as far as needed, and the army is one short every time.

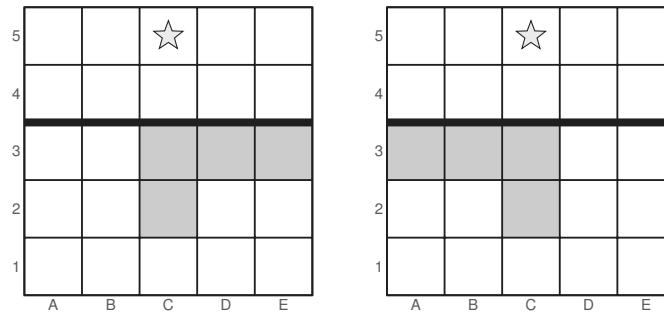
Hints, in order. “The first two values are 1 and 1.” “Each new value is the sum of the last two.”

Problem 6. Board O: C2 over C3 lands on C4.

Problem 7. Three pegs never reach C5; the highest row reached is 4 (e.g. from C2, C3 and any third square). Quick reason: each crossing uses two pegs and empties the square under the crossed peg, and the third peg cannot refill it alone; the full values reason is Problem 10.

Hints, in order. “Get one peg across first. Which square must be full under it for the last jump, and can the third peg get there?”

Problem 8. The two shapes that work are the L of C2, C3, D3, E3 (C2 over C3 to C4; E3 over D3 to C3; C3 over C4 to C5) and its mirror image C2, C3, B3, A3 (C2 over C3 to C4; A3 over B3 to C3; C3 over C4 to C5). Among all 1365 placements of four pegs on the fifteen home squares these are the only two that reach the star (computer check). Why: the last jump is C3 over C4 to C5, so C3 and C4 must be full at once; C4 can only be filled by C2 over C3 (nothing is across yet), which empties C3; so C3 must be refilled from the side, by a peg two squares away jumping over its neighbor, which is exactly the two pegs along row 3 on one side.



Shapes that fail, with the highest row they reach, for the “why not” part: the block C2 D2 C3 D3 (row 4; after one crossing the two pegs in row 4 can only move sideways); a row of four along the river (never crosses; crossing needs two pegs stacked); the T of B3 C3 D3 C2 (row 4; after C2 over C3 the remaining pegs are not adjacent); a column C1 C2 C3 with D3 (row 4); and the L itself shifted one column, such as B3 C3 D3 D2 (reaches D5, row 5 but not the star). A group that finds a shifted L has found the idea and only needs to slide it.

Hints, in order. “Which two squares must both be full just before the last jump?” “Once a peg has crossed, the square under it is empty; refill it from the side.” “Two pegs in a row, next to the stack.”

Problem 9. Board P values (rows 5 to 1): 8 13 21 13 8 / 5 8 13 8 5 / 3 5 8 5 3 / 2 3 5 3 2 / 1 2 3 2 1. (a) C2 over C3 to C4: 13 before, 13 after. (b) E3 over D3 to C3: 8 and 8. (c) B3 over C3 to D3: 13 before, 5 after. (d) C4 over C3 to C2: 21 before, 5 after. The rule: a square’s value depends on its distance from the star in steps along rows and columns (the star is 21, its four neighbors 13, and so on down the Fibonacci numbers); a jump that ends nearer the star keeps the total, any other jump lowers it.

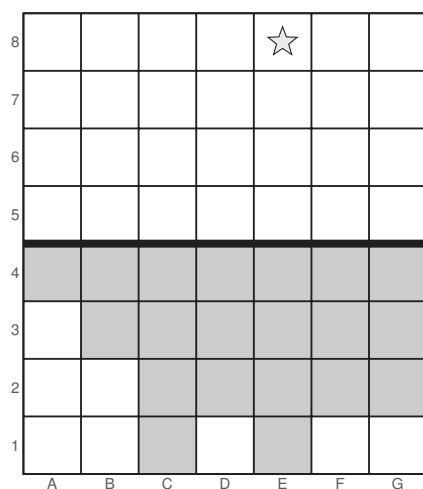
Problem 10. Best three home squares $8+5+5 = 18$; star 21; either L (C2, C3 and two of the row-3 squares) $8+5+5+3 = 21$; the block C2 D2 C3 D3: $5+3+8+5 = 21$. So three pegs, placed anywhere, are worth at most 18 and can never reach a square worth 21: Problem 7 is settled for all placements at once. The block is worth 21 and still fails (its best is row 4), so enough value does not guarantee success; the values only ever prove impossibility. The L spends its 21 exactly, each jump keeping the total, and ends with one peg on the star.

Problem 11. Board Q: C3 over C4 to C5; C1 over C2 to C3; A4 over B4 to C4; C4 over C5 to C6; E4 over D4 to C4; C3 over C4 to C5; C5 over C6 to C7. A group that plans backward (“C7 needs C6 and C5 full; C6 needs C5 and C4 full . . .”) finds it in a few minutes; a group jumping at random may need the first hint.

Hints, in order. “Plan the last jump first.” “The column of three under the middle is there to fill C5 twice.” “The two ends of the row of five are there to fill C4 twice.”

Problem 12. Board R values (rows 7 to 1): 21 34 55 34 21 / 13 21 34 21 13 / 8 13 21 13 8 / 5 8 13 8 5 / 3 5 8 5 3 / 2 3 5 3 2 / 1 2 3 2 1. Seven most valuable home squares: $13+8+8+8+5+5+5 = 52$; star 55; the T of board Q: $13+8+8+5+5+8+5+3 = 55$. So seven pegs never reach C7 (52 is less than 55, and the total never rises), and any eight-peg army that does must be worth at least 55 and spend it exactly.

Problem 13. Board T (columns A–G, rows 1–8, river above row 4, star at E8). The board is deliberately wider on the left of the star (four columns to its left, two to its right): the army needs the room on that side, so nobody should look for a misprint. Twenty pegs: A4 B4 C4 D4 E4 F4 G4 (row 4); B3 C3 D3 E3 F3 G3 (row 3); C2 D2 E2 F2 G2 (row 2); C1 E1 (row 1). The last jump is E6 over E7 to E8, so E6 and E7 must both be full at the end; that is what the “last jump” line on the page is for. Nineteen jumps:



1. G3 over G4 to G5
2. F3 over F4 to F5
3. G5 over F5 to E5
4. E4 over E5 to E6
5. E2 over E3 to E4
6. G2 over F2 to E2
7. E1 over E2 to E3
8. E3 over E4 to E5
9. E5 over E6 to E7
10. D3 over D4 to D5
11. C3 over C4 to C5
12. C5 over D5 to E5
13. C1 over C2 to C3
14. B3 over C3 to D3
15. D2 over D3 to D4
16. A4 over B4 to C4
17. C4 over D4 to E4
18. E4 over E5 to E6
19. E6 over E7 to E8

The structure is worth pointing out: jumps 1–9 use only the ten pegs in columns E–G (a three-by-three block plus E1) and park a peg on E7; jumps 10–12 bring a peg from the left half to E5; jumps 13–17 bring a second peg to E4 while E5 is full; jumps 18–19 finish. With Fibonacci values scaled so that the star is 377, the twenty squares are worth exactly 377, so every jump must keep the total. On board T the nineteen most valuable home squares are worth only 372 (fifteen squares within seven steps of the star, worth 340, and six squares at eight steps, worth 8 each), so the values alone show that nineteen pegs cannot reach E8 on this board. On an unbounded board nineteen pegs can be worth exactly 377 (the sixteen squares within seven steps plus three of the nine at eight steps, 84 sets); a nineteen-peg solution would have to be perfect, and an exhaustive search over those 84 sets, restricted to weight-preserving jumps, finds none, which is why twenty is the minimum (section 9).

Hints, in order. “The last jump is E6 over E7 to E8, so plan to park a peg on E7 and then bring another to E6 with E5 full.” “The right three columns and the column under E are a row-three army; use them first.” “After the peg is on E7, the left half must deliver a peg to E4 while E5 is full.”

Problem 14. The only other eight-peg sets that reach C7 are the two flags: A4 B4 C4 D4 E4 with C3 D3 E3 (E3 over E4 to E5; D3 over D4 to D5; E5 over D5 to C5; C4 over C5 to C6; A4 over B4 to C4; C3 over C4 to C5; C5 over C6 to C7) and its mirror image A4 B4 C4 D4 E4 with A3 B3 C3 (C3 over C4 to C5; E4 over D4 to C4; C4 over C5 to C6; B3 over B4 to B5; A3 over A4 to A5; A5 over B5 to C5; C5 over C6 to C7). Why nothing else: an eight-square set worth at least 55 must contain C4 (13) and B4, D4, C3 (8 each), and then either four of the five 5-valued squares (57 in all; none of the five such sets works) or three 5-valued squares and one of the five 3-valued home squares of the board (55 exactly; of these 50 sets, exactly the T and the two flags work). Both statements were checked by exhaustive computer search over all 125,970 sets. The four small boards on the page record the sets tried on the way.

Hints, in order. “Keep the five pegs along the river. Move the three pegs below the middle to a row along row 3, on one side (C3 D3 E3 or A3 B3 C3).” “Two of those three cross side by side; then one jumps over the other into C5.”

If the table stalls. Problems 4–5 are the heart; give them time and let the children do the sums with pennies on the strip. If the table is behind at 0:21, use only strip F1 in Problem 3 and stop Problem 8 after the first L and its mirror image. Problems 9–10 can be shortened to their first two blanks, since Problem 12 carries the same idea. If Problem 11 goes past seven minutes, give jumps 1–3. Board T (Problem 13) should start by 0:40 whatever has been skipped; Problem 14 is optional.

If the table finishes early. (a) Row five: tell them the true story (section 8) and ask what values would be needed, given that the argument must work for an army of any size; they will see that Fibonacci numbers run out of “room” on an infinite board and the mathematician can introduce the number σ with $\sigma^2 + \sigma = 1$. (b) Narrow armies: a two-wide column of any depth never passes row 1 (on a board exactly two squares wide each column is a strip on its own; on a wide board the pegs can spread sideways and the answer is still row 1: checked by search for depths up to 7 and proved for every depth in section 9); a three-wide army of any depth never reaches row 4 in any column (values: with the target above the middle column the three columns are worth $\sigma^2 + 2\sigma^3 = 3\sigma - 1 \approx 0.85 < 1$ in

the units of section 9, and less above any other column), though a three-by-four block does reach row 3, above a side column. (c) Blocks: a five-by-four block of twenty pegs reaches only row 3, a five-by-five block reaches row 4 (computer checks); ask them to find a twenty-peg shape of their own that reaches row 4.

8. The share (eight minutes)

Order: K–1, grades 2–3, grades 4–5, then the organizer. Two minutes each; children hold up pages and pieces, adults keep it moving.

- K–1 shows strip O (four pieces, the play is forced, the farthest square is 1) and board U (the L shape reaching the star in three jumps).
- Grades 2–3 show strip S with the values: the six pegs are worth 20, square 2 is worth 21, and no jump ever makes the total bigger. “The army is always one short.” And board W, if they reached it: eight pegs worth exactly 55 reaching the square worth 55.
- Grades 4–5 show the T of eight reaching row 3, and, if they got there, board T with twenty pegs reaching row 4.
- Organizer, one minute: on an endless grid with an endless army below the line, John Conway asked how high a single peg can be sent. Four rows, with twenty pegs, as just shown; and never five, not with any army, however large, in any number of moves. The proof is the same idea the tables used, values that a jump can never increase, with one special number in place of the Fibonacci numbers. Next time we can look at that number.

9. The mathematics behind the session (for the two mathematicians)

The potential. Fix the target square T and give the square at Manhattan distance d from T the weight σ^d , where $\sigma = (\sqrt{5} - 1)/2 \approx 0.618$ satisfies $\sigma^2 + \sigma = 1$. Consider a jump along three collinear squares at distances d_1, d_2, d_3 from T (jumper, jumped, landing). Either the jump moves toward T along that line, $(d + 2, d + 1, d)$, and the weight change is $\sigma^d - \sigma^{d+1} - \sigma^{d+2} = \sigma^d(1 - \sigma - \sigma^2) = 0$; or it moves away, $(d, d + 1, d + 2)$, change $\sigma^{d+2} - \sigma^d - \sigma^{d+1} < 0$; or it crosses the line through T , $(d + 1, d, d + 1)$, change $-\sigma^d < 0$. So the total weight W of the pegs on the board never increases: W is a monovariant. Reaching T needs $W \geq 1$ at that moment, so an initial position with $W < 1$ can never reach T . This is the argument of Berlekamp, Conway and Guy (*Winning Ways*, vol. 4, ch. 23, “Purging Pegs Properly”) and of Aigner’s “Moving into the desert with Fibonacci” (Math. Magazine 70, 1997); Honsberger’s *Mathematical Gems II* also has a chapter on it.

Row five is unreachable. With T in row 5, the column under T has weight $\sigma^5 + \sigma^6 + \dots = \sigma^5/(1 - \sigma) = \sigma^5/\sigma^2 = \sigma^3$, and the columns at horizontal distance $k \geq 1$ have weight σ^{3+k} each. The whole half-plane is worth $\sigma^3 + 2 \sum_{k \geq 1} \sigma^{3+k} = \sigma^3 + 2\sigma^4/\sigma^2 = \sigma^3 + 2\sigma^2$. Using $\sigma^2 = 1 - \sigma$ and $\sigma^3 = \sigma - \sigma^2 = 2\sigma - 1$, this is $2\sigma - 1 + 2 - 2\sigma = 1$: the infinite army is worth exactly the target square. A finite army is worth strictly less than 1, and W never increases, so no finite army reaches row five; and any actual play uses finitely many pegs. (For T in row 4 the half-plane is worth $1/\sigma = \phi > 1$, which is why the argument does not exclude row 4, and indeed 20 pegs suffice.)

One dimension. On a strip with T two squares past the line, the home squares are worth $\sigma^2 + \sigma^3 + \dots = 1$ exactly, and one square past the line the whole home side is worth $\sigma^{-1} > 1$. So a finite army reaches square 1 (two pegs do) and never square 2. Whether the army is contiguous or has gaps makes no difference. The Fibonacci version used on the pages is the same argument with the values $F_1, F_2, \dots$ read from the far end of a finite strip: a jump toward T replaces $F_k + F_{k+1}$ by F_{k+2} , and $\sum_{i=1}^n F_i = F_{n+2} - 1$, so a home side of n squares is worth $F_{n+2} - 1$ while square 2 is worth F_{n+2} : “always one short”. The identity is proved by induction (adding F_{n+1} to both sides turns the case n into the case $n + 1$), and the children can discover it as the pattern 12/13, 20/21, 33/34, 54/55. The same values with the target one square past the line settle which armies cross at all: on a finite strip with the square next to the line empty, the army is worth at most $F_{n+1} - 1$ against F_{n+1} for square 1 across (12 against 13 on the printed strips), so an army crosses only if that square is full from the start, which is the rule given at every

table. Integer values work on any finite board because all values are positive there; on an infinite board the integer recurrence would have to run backward into $0, 1, -1, \dots$, and the geometric σ^d is the right replacement.

Two dimensions on the pages. The value boards use F_{n-d} at distance d with n large enough that every printed square has a positive value (board P: star 21; board R: star 55; for board T take the star to be 377). Everything the pages ask is then exact arithmetic: three home squares are worth at most $8 + 5 + 5 = 18 < 21$, seven at most $13 + 8 + 8 + 8 + 5 + 5 + 5 = 52 < 55$, so 3 and 7 pegs fail for rows 2 and 3, and 4 and 8 succeed by exhibition. The minimal armies for rows 1–4 are 2, 4, 8, 20 (*Winning Ways*; Bell, Hirschberg and Guerrero-García, “The minimum size required of a solitaire army”, *INTEGERS* 7 (2007), treat this and related boards). For row 4 the potential alone gives 19 as a lower bound on the unbounded board: the nineteen heaviest home squares (the sixteen within seven steps and three of the nine at eight steps) are worth exactly 1, i.e. 377 in the units of board T, and there are $\binom{9}{3} = 84$ such sets. A nineteen-peg solution would therefore have to be *perfect*, every jump toward the target and the final position a single peg; an exhaustive search over the 84 candidate sets, restricted to weight-preserving jumps, finds none, which is how “20” was confirmed for this guide. On the printed board T the bound is already decisive: the star sits four columns from the left edge and two from the right, only fifteen home squares lie within seven steps of it (worth 340) and six at eight steps (worth 8 each), so the nineteen most valuable home squares of board T are worth $372 < 377$. The twenty-peg set on board T is worth exactly 377, so its nineteen jumps are all weight-preserving, and it ends with one peg. The exact-equality cases are the subtle part of the whole subject: for row 3, exactly three eight-peg sets work (the T and the two flags), and for row 2 exactly the two L shapes.

Sufficiency is a different matter. The block of four (worth 21) and the T of four (worth 23) cannot reach row 2; the value argument is a certificate of impossibility only. Deciding reachability in general is hard: solvability of generalized peg solitaire is NP-complete (Uehara and Iwata, 1990), which is why the positive answers on the pages come from explicit jump sequences found by search.

Pagoda functions. In peg solitaire proper (clear a board to one peg), a function p on the squares with $p(a) + p(b) \geq p(c)$ for every jump a over b to c is called a pagoda function; the total of p over the pegs never increases, so a target position worth more than the start is unreachable. Our value boards are pagoda functions for the solitaire-army problem, and the mod-3 coloring used at the grades 2–3 table (three colors, every jump flips the parity of all three counts) is the classical parity certificate of *Winning Ways*; both are necessary conditions only, as strip H and the block of four show. Beeler and Hoilman (“Peg solitaire on graphs”, *Discrete Math.* 311, 2011) start the study of the same game on arbitrary graphs.

Two-wide armies. The finish-early notes claim that a column two squares wide never sends a peg past row 1, however deep, even on a wide board where the pegs can spread sideways. On a board exactly two squares wide this is immediate (no sideways jump exists, so each column is a strip and the one-dimensional bound applies). On a wide board use a product pagoda function: give the square in column c and row r the weight $u(c)v(r)$, where $v(r) = \sigma^{2-r}$ (row 2 is the target row, rows $r \leq 0$ hold the army) and $u \equiv 1$ except $u = \varepsilon$ on the army’s right column. A vertical jump in column c changes the total by $u(c)$ times $v(r+2) - v(r) - v(r+1) = 0$ (upward) or $v(r) - v(r+1) - v(r+2) < 0$ (downward); a horizontal jump in row r from column a over b to c changes it by $v(r)$ times $u(c) - u(a) - u(b)$, which is at most $1 - (1 + \varepsilon) < 0$ since the two removed pegs stand in different columns. So the total never increases. The column of depth h is worth $(1 + \varepsilon)(\sigma^2 + \sigma^3 + \dots + \sigma^{h+1}) = (1 + \varepsilon)(1 - \sigma^h)$, which is below 1 once $\varepsilon < \sigma^h/(1 - \sigma^h)$, while every row-2 square outside the ε column is worth exactly 1; swapping the roles of the two columns covers the remaining square. Hence a two-wide army of any depth never reaches row 2 in any column, on any board. (A search on a nine-wide board confirms row 1 for depths 2 to 7.)

Facts checked by computer for this session (any of them can be handed to a fast table): all 1365 four-peg placements on the fifteen home squares of the row-2 board were searched, and only the two L shapes reach the star, while four shifted L shapes reach row 2 beside it; all 125,970 eight-peg sets on the row-3 board were searched (T and two flags); the 1-D strips of up to eight home squares were searched over every subset (21 of the 63 placements on six home squares cross, and none reaches square 2); rectangular blocks of pegs with their top row on the line reach at most the following rows, in the best column: two wide: row 1 (depths 2–7 by search on a wide board, every depth by the product weight above); three by two, three by three, four by two, four by three: row 2; three by four, four by four, five by two, five by three, five by four: row 3; five by five: row 4. A three-wide army of any

depth cannot reach row 4 in any column, by the potential: with the target above its middle column its three columns are worth $\sigma^2 + 2\sigma^3 = 3\sigma - 1 \approx 0.854$, above a side column $\sigma^2 + \sigma^3 + \sigma^4 = 2 - 2\sigma \approx 0.764$, and above any other column less still.

Where it leads. The monovariant is the basic tool of olympiad combinatorics (invariants and semi-invariants), and the same idea appears as potential functions in amortized analysis of algorithms and as Lyapunov functions in dynamics: to show a process cannot reach a state, find a quantity that only moves one way. The golden ratio enters because the jump rule forces the recurrence $v_{k+2} = v_k + v_{k+1}$; the Fibonacci strip values, Zeckendorf representations and the continued fraction of ϕ all sit next door. For a next session: the solitaire army on the triangular or hexagonal lattice and in three dimensions (the same potential with a different sum gives different maximal rows), peg solitaire on small graphs (which trees can be solved?), and the reverse question of reconfiguration: which positions can reach which.